Muhammad Taimoor Tariq

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Research Interests

My research focuses on **resource management and coordination in multi-tenant networks**, with a specific interest in designing architectures that reconcile the tension between efficiency, fairness, and performance isolation. I develop systems that enable diverse tenants with conflicting objectives to share common underlying infrastructure without degrading individual performance.

Education

University of Illinois, Urbana-Champaign **August 2022**
PhD. in Computer Science *Urbana, IL*
- Advised by: Radhika Mittal

Lahore University of Management Sciences (LUMS) **September 2018 – May 2022**
Bachelor of Science in Computer Science *Lahore, Pakistan*
- Placed on the Dean's Honors List for academic excellence for years 2020-2022

Publications

Saving Private Cellular: Addressing The Coordination Vaccum in CBRS
Taimoor Tariq, *Radhika Mittal*
HotMobile 2026

Performance Isolation for 5G RAN Slicing Across Multiple Interfering Cells
Taimoor Tariq, *Yongzhou Chen, Haitham Hassanieh, Radhika Mittal*
NINeS 2026

Centralized Traffic Engineering for Networked Farm Applications
Ammar Tahir, Yueshen Li, Jianli Jin, Changxin Zhang, Daniel Moon, Aganze Mihigo, ***Taimoor Tariq***, *Deepak Vasisht, Radhika Mittal*
IEEE / ACM SEC 2025

Enabling Emerging Edge Applications Through a 5G Control Plane Intervention
Mukhtiar Ahmad, Muhammad Nawazish Ali, ***Taimoor Tariq***, *Basit Iqbal, Taqi Raza, Zafar Ayyub Qazi*
ACM CoNEXT 2022

Neutrino: A Fast and Consistent Edge-based Cellular Control Plane
Mukhtiar Ahmad, Muhammad Nawazish Ali, ***Taimoor Tariq***, *Syed Usman Jafri, Adnan Abbass, Syeda Mashal Abbas, Basit Iqbal, Zartash Uzmi, Zafar Ayyub Qazi*
IEEE/ACM Transactions on Networking, 2022

Fast Failure Detectors for 5G Edge Deployments (Poster)
Taimoor Tariq, *Maha Kamal, Vafa Batool*
Student Research Competition, SOSP 2021

Industry Experience

Google **May 2025 – August 2025**
PhD Software Engineering Intern – Hosts: Yingjie Bi, Hamid Bazzaz *Sunnyvale, CA*
- Designed a simulation framework to estimate reservable bandwidth for Premium Bandwidth in datacenter clusters.

Teaching Experience

Teaching Assistant **August 2024 – December 2024**
CS-425: Distributed Systems *UIUC, IL*

Teaching Assistant **January 2022 – May 2022**
CS-678: Topics in Internet Research (Graduate) *LUMS, Pakistan*

Lead Teaching Assistant **September 2021 – December 2021**
CS-582: Distributed Systems (Graduate) *LUMS, Pakistan*

Teaching Assistant **January 2021 – May 2021**
CS-382: Network-Centric Computing (Undergraduate) *LUMS, Pakistan*